KEWEI LIU

M.S. Student | Multimodal AI and Speech Processing

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EDUCATION

The University of Tokyo Apr. 2026 – Mar. 2028 expected

M.S., Electrical Engineering and Information Systems, Graduate School of Engineering Tokyo, Japan

Minematsu–Saito Laboratory. Current research on AI-assisted speaking support for Japanese learners.

Kyoto University Apr. 2022 – Mar. 2026

B.Eng., Electrical and Electronic Engineering, Faculty of Engineering Kyoto, Japan

Yoshii Laboratory. Bachelor's thesis on user-controlled multimodal video summarization.

RESEARCH & PRODUCT DEVELOPMENT

User-Controlled Video Summarization First author | APSIPA ASC 2026 accepted

- Defined a video-summarization task in which users specify how the emotional tone should evolve while the summary still retains key scenes.
- Designed a multimodal model that adjusts image, audio, and subtitle contributions scene by scene and selects a sequence that balances the requested progression with content importance.
- On TVSum (50 YouTube videos, 5-fold cross-validation), improved internal trajectory alignment by 24% (0.427 to 0.531) while maintaining comparable overlap with human-rated key scenes. Led problem formulation, implementation, evaluation, analysis, and manuscript preparation.

Multidimensional Japanese Pronunciation Evaluation R&D Speech evaluation | Current R&D

- Built speech-evaluation logic for Chinese-speaking learners of Japanese, producing separate feedback on pronunciation, rhythm, fluency, and pitch.
- Diagnosed cases where mismatched or low-quality recordings received implausibly high scores and introduced a no-score policy when the available evidence was insufficient.
- Used AI coding assistants to compare implementation options and draft edge-case tests, then validated changes with audio review, structured logs, automated tests, and a browser-based failure-review tool.

PROFESSIONAL EXPERIENCE

China Tabito Education Group Shareholder · Education R&D & Product Development | Current

- Work on education research, curriculum and learning design, teaching materials, educational content, and education product development for students preparing for Japanese university admissions.
- Design physics curricula and materials for the EJU, Common Test, and university entrance examinations, informed by classroom practice and recurring learner difficulties.

Coach Academy Instructor & Admissions Advisor | 4 years

- Taught mathematics and physics to international students and advised on school selection, application planning, personal statements, interviews, and oral examinations.
- Used student questions and written errors to identify recurring barriers in reading Japanese problems, drawing physical models, and setting up equations; revised lessons and exercises accordingly.

ADDITIONAL EXPERIENCE & RECOGNITION

Kyoto University: Electrical & Electronic Circuits OA, Oct. 2024 – Mar. 2025; International Student Tutor, Apr. 2025 – Mar. 2026.

Selected internships: Sony (2026); Rakuten "Summer Internship / □□□" (2026).

Awards & Scholarships: Ichikawa International Scholarship Foundation Scholar (2023–2024); MEXT/JASSO Honors Scholarship (2022–2023); Excellence Award, Electrical and Electronic Circuits Poster Presentation, Kyoto University (2023).

TECHNICAL SKILLS & LANGUAGES

Programming	Python, PyTorch, NumPy, pandas, scikit-learn, MATLAB
ML / Evaluation	Multimodal machine learning, experiment design, cross-validation, ablation studies, statistical evaluation, failure analysis
Speech / Audio	MFCC-DTW alignment, F0, rhythm, pause, and mora-timing analysis
Development	Git/GitHub, Linux, Docker, pytest, structured logging, HTML/CSS, LaTeX
Languages	Mandarin Chinese (native), Japanese (JLPT N1), English (TOEFL iBT 81)
Interests	Music, Go (amateur 2-dan), Chinese calligraphy